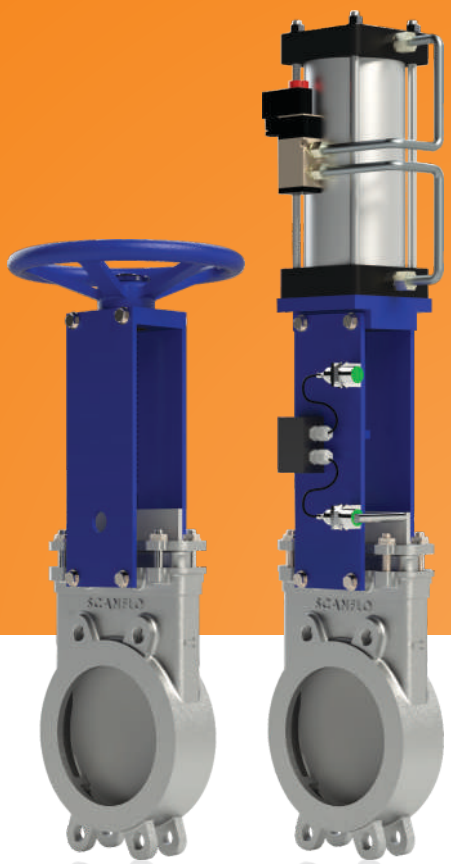
SCANFLO

FLOW RIGHT

KNIFE GATE VALVES

PRECISION ENGINEERING | SUPERIOR QUALITY





www.scanflovalves.com

INTRODUCTION

SCANFLO Valves is a wholly owned subsidiary of Scan Engineering based in Vapi, Gujarat. We specialize in designing & manufacturing of knife gate valves for toughest applications. Since established, SCANFLO has stayed true to its mission to produce quality products and to provide application based solutions in the field of valves. With over three decades of experience backed by state-of-the-art machining facility located in Gujarat & an in house foundry, we are constantly meeting customer requirements in the most efficient manner. We work with our customers as a solution provider and not as a mere product supplier.





With competent production process from foundry to final metal finish, customers can be assured of quality materials and top-class manufacturing equipment's being used to produce a valve which offers longer service life and gives them a complete peace of mind. Our team of qualified and experienced people ensures that quality is maintained at each stage and the final product meets international standards. Each valve undergoes stringent quality checks before dispatch.

Today Scanflo valves has global presence and its quality and service is highly appreciated by the customers. Our repeat orders itself speaks about the quality we offer.

VISION

SCANFLO Valves envisions global leadership in innovative valve solutions. Specializing in knife gate valves, we are dedicated to setting the industry standard for quality and durability. Our streamlined production process, coupled with a team of experts, guarantees top-class materials and manufacturing. We aspire to be the globally preferred choice, evident through repeat orders and customer acclaim."

MISSION

At SCANFLO Valves we specialize in crafting resilient knife gate valves for the most challenging applications. Our unwavering mission is to consistently deliver high-quality products and tailored valve solutions.

WHY CHOOSE US ?

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We know there are a lot of valve manufacturing firms and it is difficult to select the best partner. If you are looking for a solution provider who understands flow regulation of high consistency material like pulp, sludge, slurry etc and can provide economical solutions then we will not disappoint you.

- Its our attention to the small stuff, scheduling of timelines and high quality product that makes us stand out from the rest. We are creative, while keeping and eye on the calendar and your budget.
- With more than three decades of experience and a core technical team to assist, customers can bank on us for trouble free, peace of mind operation.
- In order to cater to our growing customer base we maintain a large amount of stock of regular valve sizes with us. This reduces your inventory carrying cost.
- On time & within budget.

We believe in delivering what we commit. We are clear from the initial proposal to set expectations and ensure success. Every transaction with our customer is to be on time, within budget and without surprises.



FOUNDRY

We cast our own valve bodies and other components. Keeping a check on the quality of the castings along with an eye on the cost.



RAW MATERIAL

We use quality stainless steel of Jindal Steel Ltd., Tata steel or Arcelor Mittal. Quality products with long lasting service life can be achieved only if the right material are used.



DESIGNING

Our designing strength is the heartbeat of our success, driven by a team of skilled engineers and designers. Meticulously crafting valves that exceed industry standards, we blend innovation with functionality. From CAD models to the final product, our commitment to precision and efficiency ensures that every valve is a testament to our designing prowess.



MACHINING

We machine almost 95% of all the valve components in-house thereby having full control on the accuracy, finishing & workmanship. Our state-of-the-art machine shop runs 24 hrs x 7 days to keep up to the increasing demands.



QC/QA DEPT

All valve parts are inspected at every stage of manufacturing so that when all the parts come together they fit in perfectly and deliver a product with highest service life

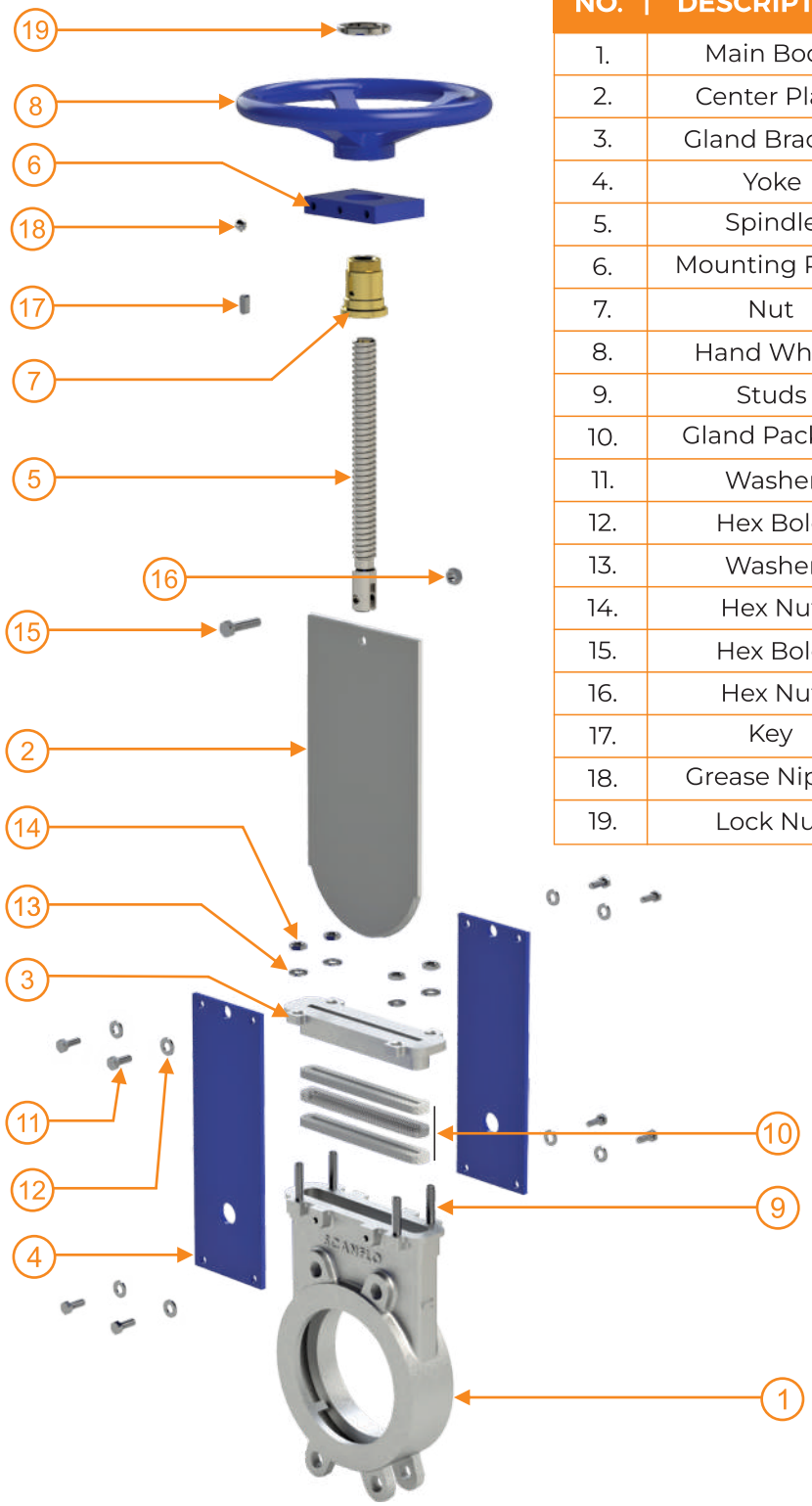


TESTING

Each valve undergoes stringent quality checks as prescribed under MSS SP 81 standards. They are fit and forget products.

WHAT MAKES A SCANFLO VALVE ?

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NO.	DESCRIPTION	MATERIAL
1.	Main Body	SS304
2.	Center Plate	SS304
3.	Gland Bracket	SS304
4.	Yoke	Carbon Steel/SS410/SS304
5.	Spindle	SS304
6.	Mounting Plate	Carbon Steel
7.	Nut	Forged Brass
8.	Hand Wheel	SG Iron
9.	Studs	SS304
10.	Gland Packing	PTFE/Carbon Filled PTFE
11.	Washer	SS304
12.	Hex Bold	SS304
13.	Washer	SS304
14.	Hex Nut	SS304
15.	Hex Bold	SS304
16.	Hex Nut	SS304
17.	Key	SS304
18.	Grease Nipple	SS304
19.	Lock Nut	Standard

DESIGN & CONSTRUCTION

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Scanflo knife gate Valves are designed for on-off & throttling applications. They feature a investment cast body in SS 304 & SS316. These valves are available in sizes ranging from 2" to 24" (DN 50 mm to DN 500 mm) and in metal and resilient seated versions. A variety of actuator options are available including handwheel and pneumatic cylinder. A full complement of accessories is also offered.

BODY

Reinforced monoblock investment casted body. The body provides a full continuous flow. This means that it produces no cavities in open position, so there is no turbulence in the fluid and load loss is minimal. The Heavy duty, Investment cast body provides abrasion resistance on the toughest liquid, slurry and dry material handling applications. The cast body, chest & packing chamber provides durability and integrity for all wetted parts.

STAINLESS STEEL KNIFE

The Knife is precisely grounded and machined to shear the flow of media passing through the valve. The SS304 knife provides optimum performance and life throughout its life. Our valves have thicker, stronger knives, to prevent deflection under heavy pressure and to ensure a tight seal.

STAINLESS STEEL STEM

All valves are furnished with a stainless-steel stem that assures easy operation & dependable service. The stem moves in a PB nut fitted in the mounting plate of the valve. The contact area between the nut & stem can be externally lubricated through a grease nipple. So our valves never get jammed if not operated for longer period of time.

RESILIENT SEAT & METAL SEAT

Resilient seated valves are available for applications where drip tight shut off is required. Unlike other valves SCAN Valves have a resilient seat designed especially for knife gate valve application. The resilient seat material is bonded into the stainless-steel seat ring when the valve is closed, the gate is pushed against the seat and held in place by the gate jams. When the valve is open, gate valve moves away from the seat allowing operating clearance which prevents seat damage and makes operations easy.

SEAT MATERIAL	MAX TEMP
EPDM (E)	120
Nitril (N)	100
Vitron (V)	180
PTFE with o-ring nitril (P)	100
PTFE with o-ring Viton (PV)	180
Polyurethane (U)	90

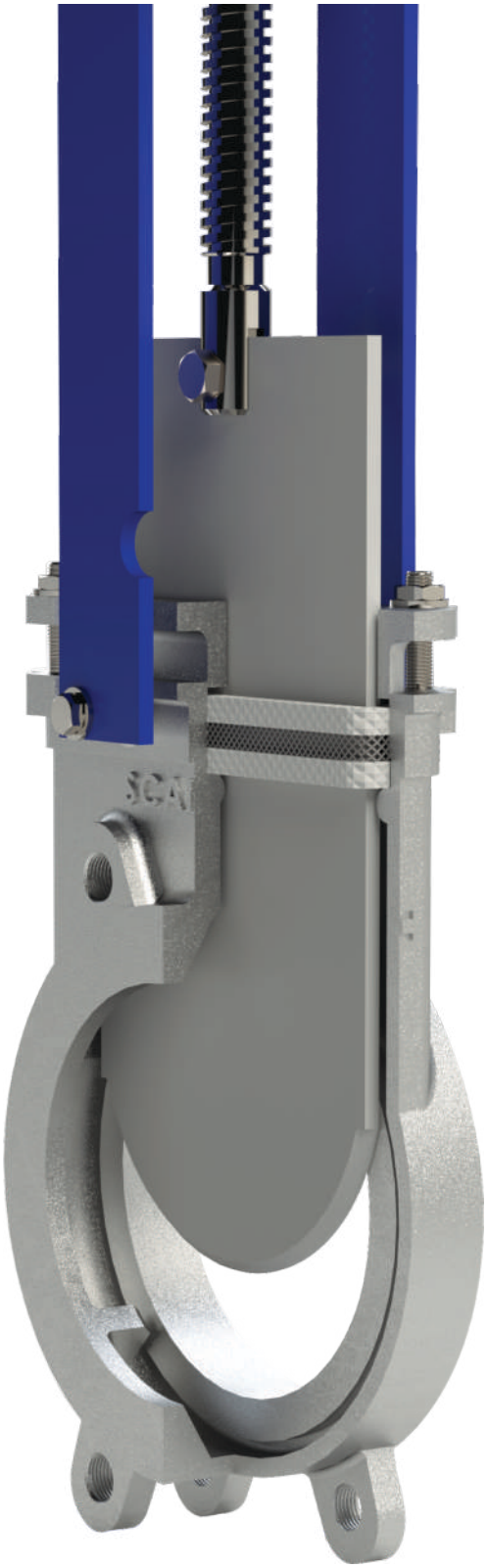
GLAND PACKING

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Standard packing in Scanflo Valves consists of a specially designed PTFE seal which provides tightness between the body and the knife gate, preventing any type of discharge to the atmosphere. It also has greased packing to help valve operation during the opening and closing functions. It is located in an easily accessible place, and can be replaced without dismantling the valve from the pipeline.

MAXIMUM TEMPERATURE RANGE

Pure PTFE	500°F 260°C	
PTFE impregnated Synthetic Fiber	450°F 232°C	
PTFE w/Buna-N Quad Seal[1]	194°F 90°C	
PTFE w/Viton® Quad Seal[1]	392°F 200°C	
PTFE w/EPDM Quad Seal	248°F 120°C	
PTFE w/Copper Wiper[2]	500°F 260°C	
Graphite	1202°F 650°C	



ACTUATOR OPTIONS

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MANUAL DRIVES

- Hand wheel
- Chain Handwheel
- Geared Motor
- Others (Square Stem)

AUTOMATIC DRIVES

- Electric Actuator
- D/A Pneumatic Cylinder
- Hydraulic Cylinder

Available in rising and non-rising stem versions.

AVAILABILITY OF ACCESSORIES

- Mechanical Stoppers
- Locking Devices
- Emergency Manual Drives
- Electrovalves
- Positioners
- Limit Switches
- Proximity Detectors
- Straight Floor Stand
- Leaning Floor Stand



Straight Floor Stand



Leaning Floor Stand

Stem extensions have also been developed, allowing the drive to be located far away from the valve, to suit all needs, please ask our engineers beforehand.



Handwheel Rising Stem



Chain Handwheel



Handwheel Gear-Box



Pneumatic Actuator



Electric-Motor Actuator



Hydraulic Actuator

SA UNI-DIRECTIONAL VALVE

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The SA model knife gate valve is a Uni directional wafer type valve designed for general industrial service applications. The knife gate fits tightly with seat and the movements between gate & seat offers shear cutting forces and self-cleaning function which makes it suitable for pulp & medium containing fiber.

FEATURES

- One piece body design with internal cast in gate wedges and guides for tighter shut off.
- Full port design for greater flow capacity & minimal pressure drop.
- Stainless steel gate grinded on both sides to avoid jamming and seat damage.
- Standard metal to metal seat (Optional: metal to soft seated).
- O' port as standard. Other options: V port, special materials, fabricated valves.

SIZE RANGE

2” (50 NB) to 28” (700 NB)

ACTUATOR OPTIONS

Manual handwheel, Pneumatic, Bevel Gearbox, Hand lever, Hydraulic, Electric, Chainwheel.

FLANGE STANDARD

ASME B16.5 Class 150, Table E' & D', DIN PN 10, DIN PN16

Part	Material of Construction
Body	CF8, CF8M
Gate	SS304, SS316
Seat	Metal Seat / Soft Seat
Packing	PTFE Impregnated Fibre
Gland Follower	CF8, CF8M
Stem	SS304
Stem Nut	Forged PB
Yoke	Carbon Steel
Seat Ring	SS304, SS316
Actuation	Manual
Handwheel	SG Iron
Stem Cover	Carbon Steel



SB BI-DIRECTIONAL VALVE

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The SB model provides repeatable bi-directional shutoff in a rugged, single piece body casting. It features a replaceable elastomer seal.

FEATURES

- One piece body / Two piece body design with different seat options.
- Standard PTFE packing provides exceptional gland sealing.
- Elastomer seals guide the plate and offers tight shut off.
- Unrestricted flow path with minimal pressure drop achieved through full port valve design.
- Deflection resistant gate design ensures consistent alignment for industry leading durability.
- Actuation accessories: Limit & proximity switches, positioners, solenoid valves, manual overrides, locking devices, fail safe systems & stem extensions.

SIZE RANGE

2" (50 NB) to 28" (700 NB)

ACTUATOR OPTION

Manual handwheel, Bevel Gearbox, Hand lever, Hydraulic, Electric, Chainwheel

FLANGE STANDARD

ASME B16.5 Class 150, Table E' & D', DIN PN 10, DIN PN16

Part	Material of Construction
Body	CF8, CF8M
Gate	SS304, SS316
Seat	Metal Seat / Soft Seat
Packing	PTFE Impregnated Fibre
Gland Follower	CF8, CF8M
Stem	SS304
Stem Nut	Forged PB
Yoke	Carbon Steel
Seat Ring	SS304, SS316
Actuation	Manual
Handwheel	SG Iron
Stem Cover	Carbon Steel



SP UNI-DIRECTIONAL VALVE (PNEUMATIC)

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The SP model pneumatic operated knife gate valve provides repeatable on-off actuation in a rugged, one-piece body casting. It features a replaceable elastomer cord for enhanced sealing. The SB model is easily automated for on/off applications.

FEATURES

- One piece body design with internal cast in gate wedg es and guides for tighter shut off.
- Full port design for greater flow capacity & minimal pressure drop.
- Stainless steel gate grinded on both sides to avoid jamming and seat damage.
- Standard metal to metal seat (Optional: metal to soft seated).
- 'O' port as standard. Other options: V port, special ma terials, fabricated valves.

SIZE RANGE

2" (50 NB) to 28" (700 NB)

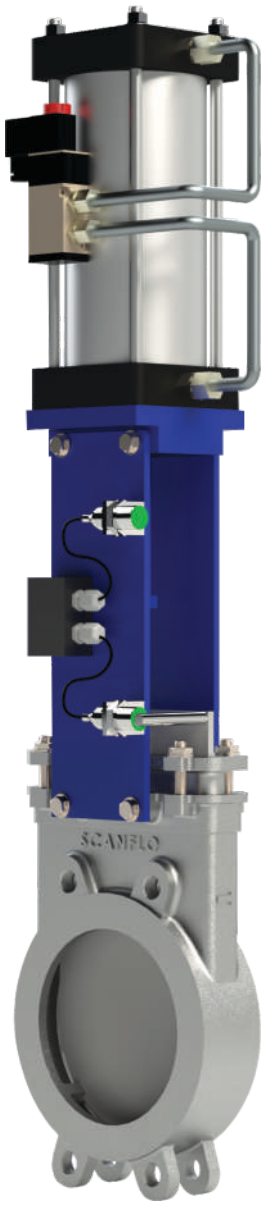
ACTUATOR OPTION

Manual handwheel, Pneumatic, Bevel Gearbox, Hand lever, Hydraulic, Electric, Chainwheel.

FLANGE STANDARD

ASME B16.5 Class 150, Table E' & D', DIN PN 10, DIN PN16

PART	MATERIAL OF CONSTRUCTION
Body	CF8, CF8M
Gate	SS304, SS316
Seat	Metal Seat / Soft Seat
Packing	PTFE Impregnated Fibre
Gland Follower	CF8, CF8M
Stem	SS304
Stem Nut	Forged PB
Yoke	Carbon Steel
Seat Ring	SS304, SS316
Actuation	Pneumatic
Cylinder	Scanflo / Duncan / Festo
Cylinder Type	Double Acting



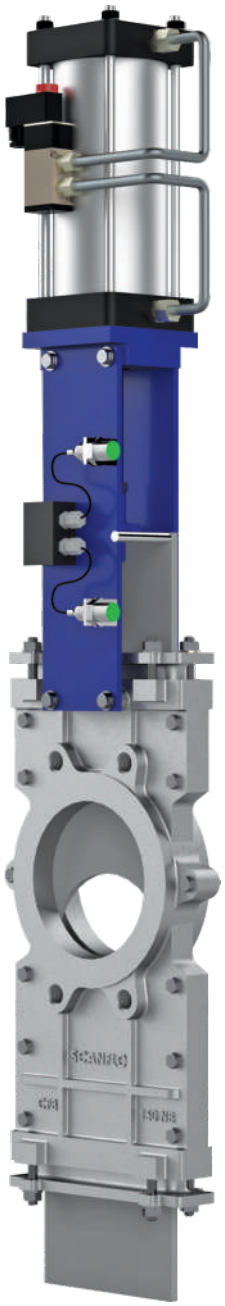
ST LONG PLATE VALVE

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The ST model knife gate valve is a through conduit wafer valve designed for media with high consistency. The double seat design assures a non-clogging shut off on either normal or reverse flow.

FEATURES

- The valve is specifically designed with no cavities to prevent stock build up.
- With an elastomer seal, bubble tight sealing is achieved in both direction.
- Full port design minimizes pressure drop & maximizes flow capacity.
- Packing at both top & bottom eliminates leakages to surrounding.
- Optional seat retainer disc design.
- With seats on both sides, valve can handle flow in both directions.



SIZE RANGE

2” (50 NB) to 28” (700 NB)

ACTUATOR OPTION

Manual handwheel, Bevel Gearbox, Hand lever,
Hydraulic, Electric, Chainwheel

FLANGE STANDARD

ASME B16.5 Class 150, Table E' & D', DIN PN 10, DIN PN16

PART	MATERIAL OF CONSTRUCTION
Body	CF8, CF8M
Gate	SS304, SS316
Seat	Metal Seat / Soft Seat
Packing	PTFE Impregnated Fibre
Gland Follower	CF8, CF8M
Stem	SS304
Stem Nut	Forged PB
Yoke	Carbon Steel
Seat Ring	SS304, SS316
Actuation	Pneumatic
Cylinder	Scanflo / Duncan / Festo
Cylinder Type	Double Acting

SD DOUBLE GATE VALVE

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The SD model knife gate valve is a double gate full lugged valve widely used in the pulp & paper industry (recycled or secondary fibre processing) designed to handle concentrated or contaminated media. In open position, both gates retract into the body & allow full flow. When closing, the gates push stock and contaminants as staples, wires etc out of the body into the flow. All components subject to wear can be replaced easily.

FEATURES

- Internally machined cast or fabricated two part body with HMWPE liners for optimum gate guiding and sliding when traveling
- Stroke time is the half of conventional valves.
- Full port design minimizes pressure drop & maximizes flow capacity
- Two cast stainless steel rings, easily replaceable, that protects the body
- Double packing with easy access and adjustable packing glands ensuring tight seal. Long life PTFE impregnated synthetic fibre & EPDM o ring
- Actuated accessories : limit & proximity switches, mechanical stops, positioners, solenoid valves, manual overrides, locking devices & fail safe systems.



Part	Material of Construction
Body	CF8, CF8M
Gate	SS304, SS316
Seat	Metal Seat / Soft Seat
Packing	PTFE Impregnated Fibre
Gland Follower	CF8, CF8M
Stem	SS304
Stem Nut	Forged PB
Yoke	Carbon Steel
Seat Ring	SS304, SS316
Actuation	Pneumatic
Cylinder	Scanflo / Duncan / Festo
Cylinder Type	Double Acting

ACCESSORIES AND OPTIONS

Different types of accessories are available to adapt the valve to specifi c working conditions such as:

MIRROR-POLISHED KNIFE GATE: The mirror-polished knife gate is especially recommended in the food industry and, in general, in applications in which solids can stick to the gate.

PTFE-LINED KNIFE GATE: As with the mirror-polished knife gate, it improves the valve's resistance to products that can stick to the gate.

STELLITED KNIFE GATE: Stellite is added to the knife gate's lower edge to protect it from abrasion.

SCRAPER IN THE PACKING: Its function is to clean the knife gate during the opening movement and prevent possible damage to the packing.

AIR INJECTIONS IN THE PACKING: By injecting air in the packing, an air chamber is created which improves tightness.

DRIVE OR YOKE SUPPORT: Made of EPOXY-coated steel (or stainless steel to order), its robust design gives it great rigidity in order to withstand the most adverse operation conditions.

MECHANICAL LIMIT SWITCHES, INDUCTIVE DETECTORS AND POSITIONERS: Limit switches or detectors to indicate specific valve position, and positioners to indicate continuous position.

ELECTROVALVES: For air distribution to pneumatic drives.

JUNCTION BOXES, PNEUMATIC PIPING AND WIRING: Fully assembled units can be supplied with all the necessary accessories.

MECHANICAL STROKE LIMITING STOP (MECHANICAL STOPPERS): These allow the stroke to be mechanically adjusted, limiting the valve run.

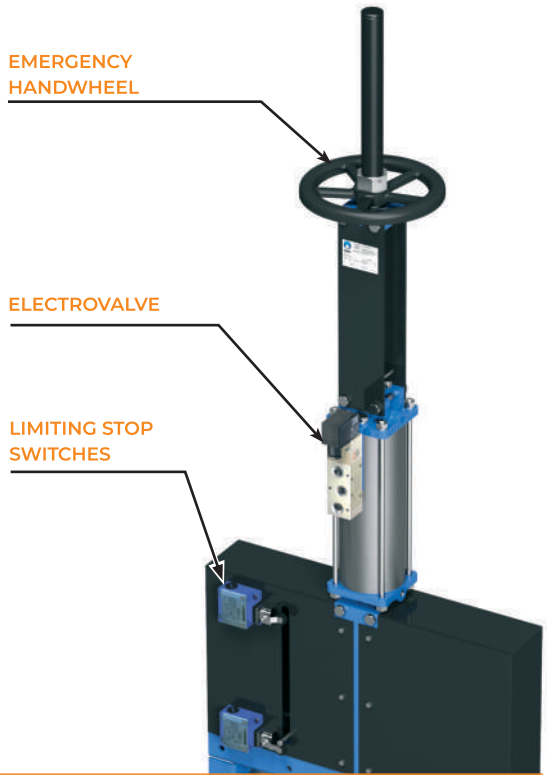
MECHANICAL LOCKING DEVICE: Allows the valve to be mechanically locked in a fixed position.

EMERGENCY MANUAL DRIVE (HANDWHEEL / GEARED MOTOR): Allows manual operation of the valve in the event of power or air failure.

INTERCHANGEABILITY OF THE DRIVES: The drives are easily interchangeable.

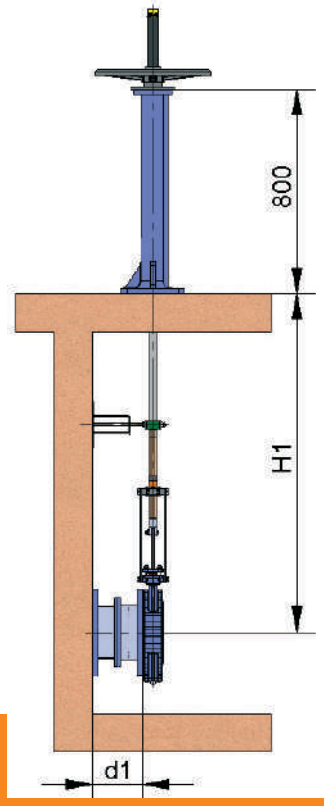
EPOXY COATING: All carbon steel valve bodies and components are EPOXY coated, making the valves hugely resistant to corrosion while giving them an excellent finish. The standard colour in **Scanflo Valves** is blue, RAL-5002.

SAFETY GUARDS FOR THE KNIFE GATE: In accordance with European Safety Standards ("EC" marking), automatic valves are equipped with metal guards for protection, preventing any objects from being accidentally caught or dragged along in the knife gate's run.



TYPES OF EXTENSIONS

When the valve needs to be operated from a distance, the following types of drives can be fitted:



STANDARD OPERATION STAND

1- FLOOR STAND

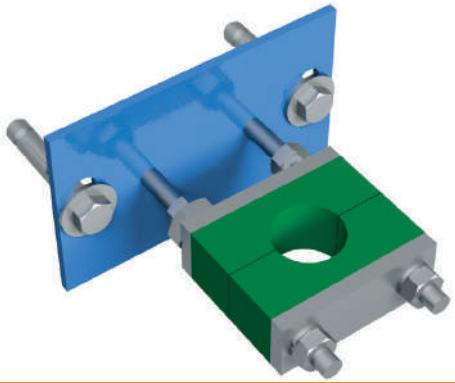
This extension is done by coupling a spindle to the stem. The desired extension is achieved by defining the length of the spindle. A floor stand is normally installed to support the drive.

The definition variables are as follows:

H1 =	Distance from valve centre to base of the stand
D1 =	Separation from the wall to the end of the connecting fl ange

CHARACTERISTICS

- It can be coupled to any type of drive.
- We recommend a stem guide bracket every 1.5 m
- The standard floor stand is 800 mm high.
- Option to use a position indicator to determine the valve's percentage of opening.
- Leaning stand available to order .
- Other floor stand measurements available on request.

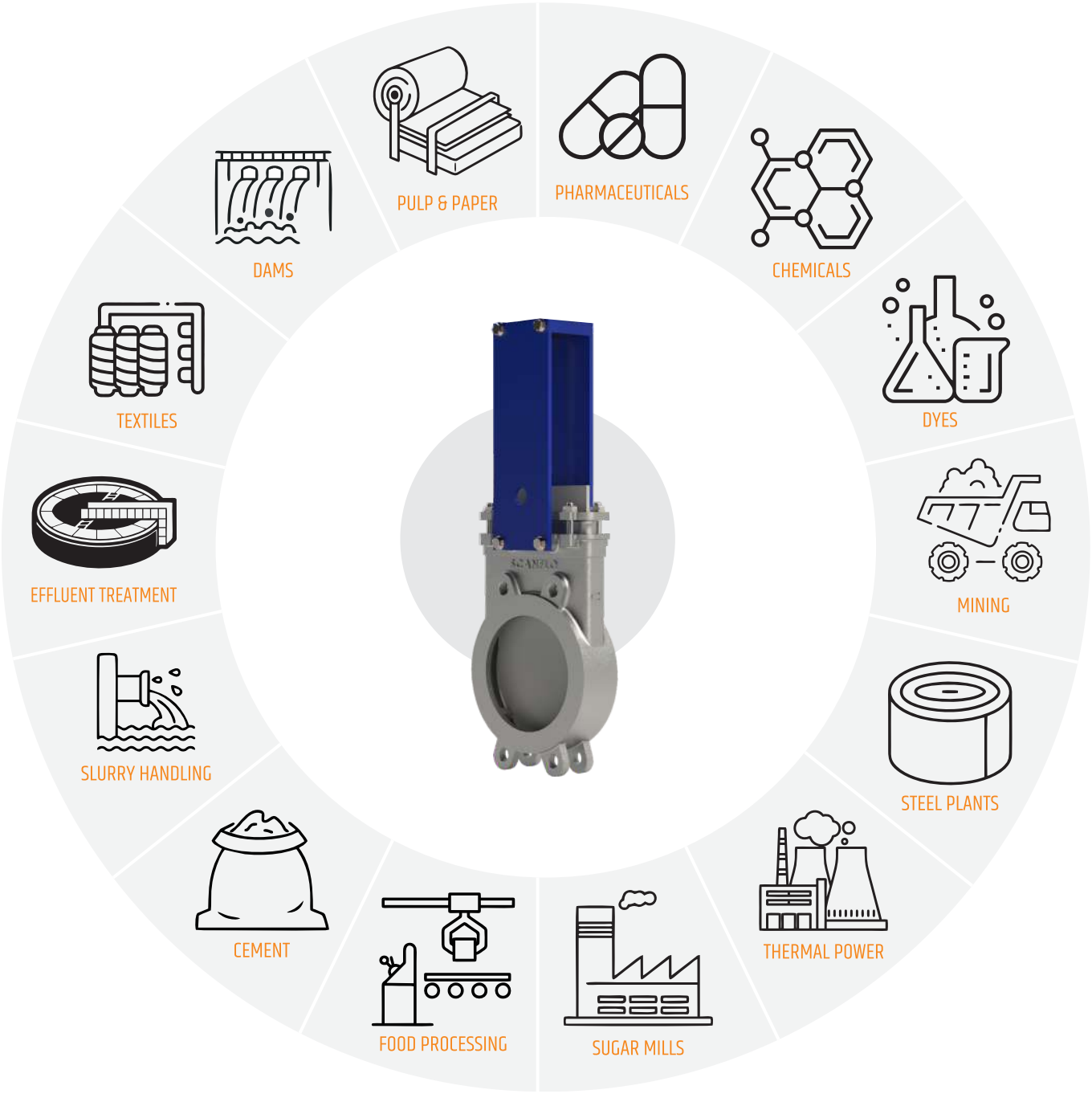


STEM GUIDE BRACKET



APPLICATIONS

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